Taller 3 circuito 2

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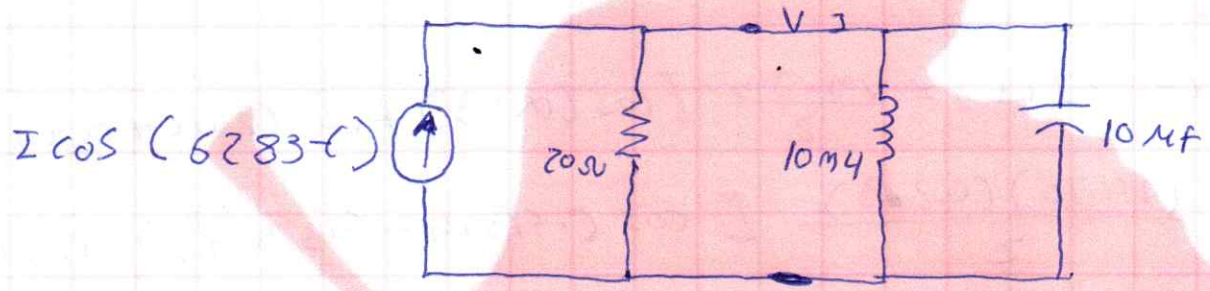
1) valor de Potencias istananeas del inductor

• $\omega = 6283$ [rad/s] • $I_s = 1 \cdot \cos(\omega t)$ A



$Z_R = 20\Omega$

$Z_L = j\omega L = j6283\,(0{,}01)$

$Z_L = 62.83j\,\Omega$

$Z_C = \frac{1}{j\omega C} = \frac{1}{j(6283)(10\times10^{-6})} = -j15.91\,\Omega$

$I = 1\angle 0$

se Hallo corriente en Zaductor

$I_L = \frac{14.70\angle -43.19}{j62.53} = 0.23\angle -133.19$ [A]

$\frac{V_1}{20} + \frac{V_1}{62.83j} + \frac{V_1}{-j15.91} = 1\angle 0$

$V_1\, 0.068\angle 43.19 = 1\angle 0$

$V_1 = \frac{1\angle 0}{0.068\angle 43.19}$

$V_1 = 14.70\angle -43.19$ [V]

mismo voltaje en las ramas

$D(t) = V(t) \cdot I(t)$, $V1(t) = 14.70 \cos(6283t - 43.19)$

$I2(t) = 0.23 \cos(6283t - 133.19)$

$P(t) = 14.70 \cos(6283t - 43.19) \cdot (0.23 \cos(6283t - 133.19)]$

Formula

$P(t) = V(t) \cdot I(t) = \frac{Vm \cdot Im}{2} [\cos(\alpha - \beta) + \cos(2\omega t + \alpha t \beta)]$

$P(t) = \frac{(14.70)(0.23)}{2} [\cos(-43.19 - (-133.19)) + \cos(2\omega t - 43$

$- 133.19)]$

$P(t) = 1.69 [\cancel{\cos(90^\circ)} + \cos(2\omega t - 176.38]$

$P(t) = 1.69 \cos(2\omega t - 176.38)\ W$

$\downarrow$

$P = I$

potencia inductancia

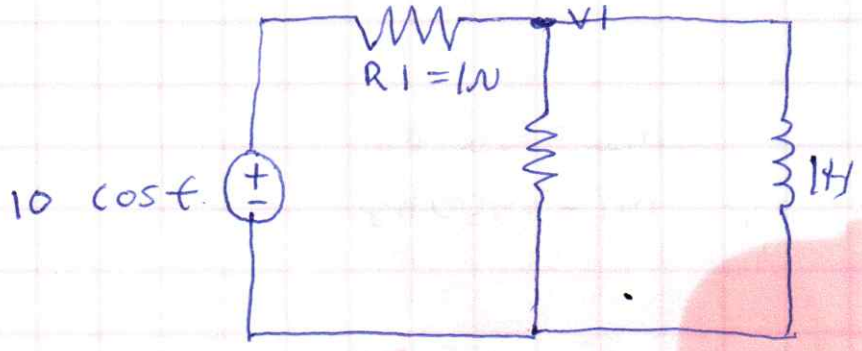
Potencia Activa o promedio

$P = \frac{1}{2} (14.70)(0.25) \cos[-43.19 - (-133.19)]$

$P = 1.69 \cancel{\cos^\circ(90^\circ)}$

$P = 0.W$

2 Potencia Actuta ó media)



• 10 cos (t) entonces $\omega = 1$

$\omega = 1 \text{ Rad/s}$

$$V1 - \frac{10\angle 0}{1} + \frac{V1}{1} + \frac{V1}{j1} = 0$$

$$\frac{V1}{1} + \frac{V1}{1} + \frac{V1}{j1} = 10\angle 0$$

$$2.23\angle -26.56\, V1 = 10\angle 0$$

$$V1 = 4.48\angle 26.56 \text{ [V]}$$

Inductancia

$$ZL = j\omega L = j(1)(1) = j1\Omega$$

Fuente $= 10\angle 0$

corriente en R1

$$IR1 = \frac{6.31\angle -18.48}{1} =$$

$$IR1 = 6.31\angle -18.48 \text{ [A]}$$

voltaje en R1

$$VR1 = 10\angle 0 - 4.48\angle 26.56$$

$$VR1 = 6.31\angle -18.48 \text{ [V]}$$

$$P = \frac{1}{2}(6.31)(6.31)\cos[-18.48-(18.48)]$$

$$P = 19.90\cos(0) = 19.90\,W \approx 20\,W$$

R1 potencia

$$Z91 = \frac{1 \cdot 1}{1+j1} = 0.70\angle 45$$ → Resultado de potencia paralela

Actuta R1 + Z91

$$Zp_0 = 1 + 0.70\angle 45 = 1.57\angle 18.31$$

Hallamos la corriente total

$$I1 = \frac{10\angle 0}{1.57\angle 18.31} = 6.36\angle -18.31$$

entonces la potencia es

$$P = \frac{1}{2}(10)(6.36)\cos[0-(-18.31)]$$

$$P = 31.8\cos(18.31)$$

$$P = 30.18\,W \approx 30\,W$$

GOD OF THE HEAVEN ALL SRIGHT WITH THE WORLD

talleres: 1,7,8 = 1,7,8 barques más 1,8
talleres: 1,3,4 =

$m1 = 950\ kg$
$m2 = 1900\ kg$

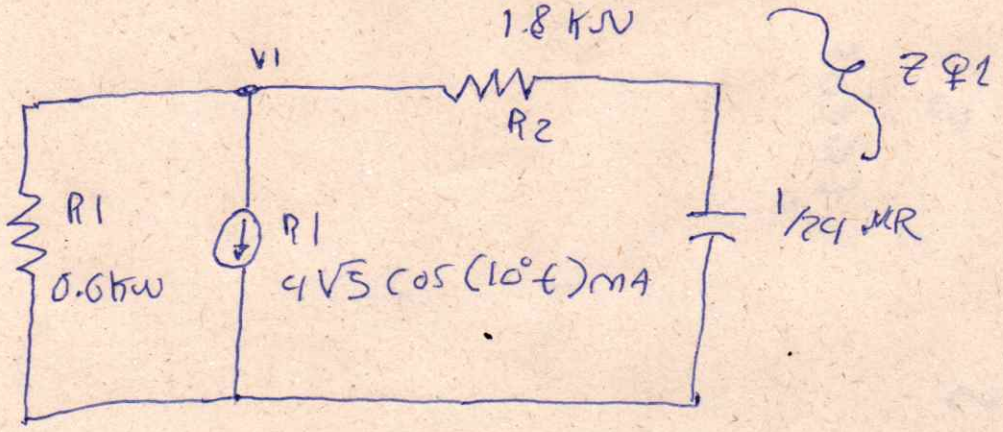
colisión perfectamente inelástica: $\vec{P_i} = \vec{P_F}$

$m_1 \vec{v_{1i}} + m_2 \vec{v_{2i}} = (m_1 + m_2) \vec{v_F}$

$950 v_{1i}\, i + 1900 v_{2i}\, j = 416$

NERD
GOD'S IN HIS HEAVEN, ALL'S RIGHT WITH THE WORLD.

b Determinar potencia



$R1 = 60$ $\omega = 10$

$R2 = 1800\Omega$

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j(10.000)(\frac{1}{24}\mu F)} = j2,4390.2\Omega$$

$Zq1 = 4390236\angle 90\Omega$

$I1 = 0.00894\angle 0$

$$\frac{V1}{600} + \frac{V1}{4390236\angle 90} = 0.00894\angle 0$$

$V1\,0.001666\angle -0.007 = -0.00894\angle 0$

$V1 = 5.36\angle -179.99$

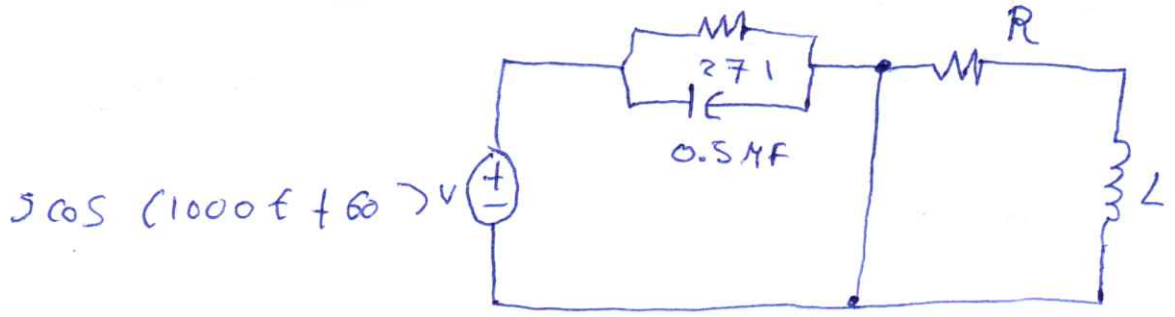
Abero

corriente IR1

$$IR1 = \frac{5.36\angle -179.99}{600}$$

$= 0.008\angle -179.99$ [A]

3 transferencia maxima de Potencia



$$Zth = \frac{4000 - j2000}{4000 - j2000} = \frac{-j8 \times 10^6}{4000 - j2000}$$

$F1 = 5 \angle 6$ V

$\omega = 1000$ rad/s

$$Zc = \frac{1}{j(1000)(0.5 \times 10^{-6})}$$

$Zc = -j2000\Omega$

$4000 + j2000$ Para obtener $a + jb$
real imaginatario

Entonces

$$Zth = \frac{-j8 \times 10^6}{4000 - j2000} - \frac{4000 + j2000}{4000 + j2000} = \frac{-j32 \times 10^9 + 16 \times 10^9}{16 \times 10^6 + j8000 - j8000 + 4000}$$

$$= \frac{-j32 \times 10^9}{20 \times 10^6} + \frac{16 \times 10^9}{20 \times 10^6}$$

Potencia Activa en R1

$$P = \frac{1}{2}(5.36)(0.008)\cos(-179.99-(-179.99))$$

$$P = 0.0244\,W \rightarrow [24.4\,mW]$$

Ahora en f1

$$P = \frac{1}{2}(5.36)(0.00844)\cos[-179.99-(0)]$$

$P = -0.023\,W \rightarrow$ potencia consumida

Ahora en :

$$I_{zq} = \frac{5.36\angle 179.99}{4390236\angle 90} = 0.00000122\angle 89.99$$

$$V_C = \frac{0.00.00122\angle 89.99}{-j2439.02} = 0.00297\angle 179.99$$

$$P = \frac{1}{2}(0.00297)(0.0000122)\cos[89.99-179.99]$$

$$P = 0\,W$$

$$z_{th} = \frac{-j32 \times 10^4}{20 \times 10^6} + \frac{16 \times 10^4}{20 \times 10^6} = 800 - j1600\,\Omega$$

$$z_{eq} = R + j\omega L$$

entonces

$R = 800$ - parte real $= R$

$z_L = -j1600$ - parte imaginaria

Para hallar L: $L = \frac{1600}{\omega} = \frac{16\cancel{00}}{10\cancel{00}} = 1.6\,H$